THE (N+1)-ST TERM OF COSK_N SK_N V AS A MAPPING OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the internal-hom device Mor_C(X, W) = ([n] |-> Mor_C(X, W_n)) and the identity Mor(U x X, W) = Mor(U, Mor_C(X, W)) are stated and used explicitly (source lines 3447-3453); the first equality is acknowledged by the author as 'obvious' (source line 3445) and is definitional for the skeletons; no verification is omitted.

Editorial note. The label lemma-cosk-shriek names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Morphism objects from coproducts of representables. Same chapter, \label{lemma-morphism-from-coproduct}. Proof lines 3426-3429: the left-hand side represents the functor X |-> Mor(X x Delta[n+1], cosk_n sk_n V).
- The cosk-sk adjunction. Same chapter, Section \label{section-coskeleton} (coskeleton constructions). Proof line 3430: Mor(X x Delta[n+1], cosk_n sk_n V) = Mor(X x sk_n Delta[n+1], sk_n V).

Source reference. Original reference 017F: sources/stacks-simplicial/pinned-original-source.tex, source lines 1292–1292. Prerequisite pointer invoked at proof line(s) 3431; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a category with finite coproducts and finite limits. Let V be a simplicial object of $\mathcal{C}$. In this case*

$$(cosk_n sk_n V)_{n+1} = \mathrm{Hom}(i_{n!} sk_n \Delta[n+1], V)_0.$$

Proof. By Lemma 017F the object on the left represents the functor which assigns to X the first set of the following equalities

$$\begin{aligned} \mathrm{Mor}(X \times \Delta[n+1], \mathrm{cosk}_n \mathrm{sk}_n V) &= \mathrm{Mor}(X \times \mathrm{sk}_n \Delta[n+1], \mathrm{sk}_n V) \\ &= \mathrm{Mor}(X \times i_{n!} \mathrm{sk}_n \Delta[n+1], V). \end{aligned}$$

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The object on the right in the formula of the lemma is represented by the functor which assigns to X the last set in the sequence of equalities. This proves the result.

In the sequence of equalities we have used that $\mathrm{sk}_n(X \times \Delta[n+1]) = X \times \mathrm{sk}_n\Delta[n+1]$ and that $i_{n!}(X \times \mathrm{sk}_n\Delta[n+1]) = X \times i_{n!}\mathrm{sk}_n\Delta[n+1]$. The first equality is obvious. For any (possibly truncated) simplicial object W of $\mathcal{C}$ and any object X of $\mathcal{C}$ denote temporarily $\mathrm{Mor}_{\mathcal{C}}(X, W)$ the (possibly truncated) simplicial set $[n] \mapsto \mathrm{Mor}_{\mathcal{C}}(X, W_n)$. From the definitions it follows that $\mathrm{Mor}(U \times X, W) = \mathrm{Mor}(U, \mathrm{Mor}_{\mathcal{C}}(X, W))$ for any (possibly truncated) simplicial set U. Hence

$$\begin{aligned} \mathrm{Mor}(X \times i_{n!}\mathrm{sk}_n\Delta[n+1], W) &= \mathrm{Mor}(i_{n!}\mathrm{sk}_n\Delta[n+1], \mathrm{Mor}_{\mathcal{C}}(X, W)) \\ &= \mathrm{Mor}(\mathrm{sk}_n\Delta[n+1], \mathrm{sk}_n \mathrm{Mor}_{\mathcal{C}}(X, W)) \\ &= \mathrm{Mor}(X \times \mathrm{sk}_n\Delta[n+1], \mathrm{sk}_n W) \\ &= \mathrm{Mor}(i_{n!}(X \times \mathrm{sk}_n\Delta[n+1]), W). \end{aligned}$$

This proves the second equality used, and ends the proof of the lemma. $\square$